

Program ω Plan: The State

Draft for hostile review — frozen upon the reviewer’s confirmation. No stage computation runs before the review completes. Companion to papers I-V (closed), the synthesis (closed), and the registry. All standing disciplines carry over unchanged: two implementations for every load-bearing number, pre-registration before computation, tag discipline in prose, the diff protocol for documentary work, instruments defined before values are seen (P6), reproduce-before-explain, the falsifier-visibility check, and the error ledger as part of every record.

0. Charter

The chirality arc ended by handing everything undecided to one address: the state. The orientation is an initial condition riding an anchor (V); the anchor-supplier, if any, lives outside the audited symplectic jurisdiction (REG-13); the settling mechanism’s non-Bogoliubov character is proven in scoped form with one road unclosed (REG-14); the e_R boundary’s best candidate resolution (the second wall) is a question about the scale line’s state-governed profile; and the measurement stratum long ago placed the relational, gravity-flavored character of the framework in the folium. Program ω asks, in frozen order: **what does the corpus already assert about the state (A); does the scale line carry a second wall (B); can a settling dynamics be concretely constructed and does it move depth (C); and does anything in the state stratum supply the anchor, or is the initial-condition verdict terminal (D)?**

Scope limits, fixed now. No spacetime claim anywhere; the folium’s gravitational character may be *observed* where it appears but is not developed; Reading B stands; Track P untouched; generation structure enters only if a stage forces it. Roles unchanged: TOM builds, Fable-5 reviews and independently reimplements; either side’s hand-derivations carry their flags.

1. Stage A — The ω -dossier (documentary; the state’s own checklist)

Precedent: the $\mathbb{O}e_s$ necessity audit, which found C1 on its first pass. Both sides **independently** compile everything the corpus asserts about states — the quasi-free family and χ (Foundation), the settling ledger and its scoped theorem (Foundation §[LEDGER], V §4), thermality/KMS and the derived flow’s stationarity, events and the relational/measurement structure (measurement paper) *and, separately, the folium* (Becoming and companion documents — the two were originally conflated under one citation here; corrected during Stage A’s dossier compilation, which found “folium” absent from the measurement

paper entirely), the vacua manifold and mixed endpoint (V §3), the carrier verdict and the anchor (V §5), the wall profile $m(u)$'s provenance (II/III), and the type-III citations — then **diff the dossiers**. Deliverable: one reference document with every claim tagged and cited; every contradiction or gap between strata written months apart is a finding. Completeness criterion: review target 3.

2. Stage B — The second wall (the program's agreed first physics assignment; the kinematic/dynamic split itself first proposed in `memory_note_second_wall.md`, filed before this plan existed and uncited in it until now — same split, same reasoning, no result asserted there, so B1 below is genuinely open, not a rerun)

B1 (kinematic reconnaissance; bounded; no ω -machinery). Against the closed record only: characterize the scale line's two asymptotics (the $\mu \rightarrow 0$ contraction end; the opposite end as the record has it) and determine whether the structures already derived **force, admit, or exclude** a second zero-crossing of the mass profile. Pre-priced outcomes: *forced* (the second wall exists structurally — the e_R boundary converts to geography; the mirror-wall content theorem becomes B1's immediate successor); *excluded* (a no-go worth its own note; e_R joins the named boundaries); *undetermined-without-dynamics* (the honest expected case per the reviewer's separability warning — B2 activates inside Stage C, not before).

B2 (dynamic; gated on C). The profile question asked of an actual settling candidate: does the constructed dynamics generate a profile with one crossing or two? Not scheduled until C supplies a candidate.

Standing payoff, gated and named: if the second wall lands, Yukawa-as-overlap enters the registry as a derived-mechanism candidate (L-wall/R-wall overlap through the state-governed bulk). Nothing in this plan asserts it.

3. Stage C — Constructing a settling candidate (the program's core)

Framing, fixed in advance to keep the corpus honest: Becoming's ledger states no settling mechanism is currently supplied; V proved the specified class is a carrier. Stage C therefore performs **candidate-construction, not derivation** — building a concrete dynamics from the framework's own materials and testing it, with every result tagged as a property of the candidate, never of the framework, unless uniqueness is separately proven.

C1 — the preservation lemma (REG-14), a genuine proof attempt. Unitaries mapping the quasi-free family into itself are Bogoliubov: attempt the proof (or locate and verify it in the literature per the cite-and-verify standard).

Success upgrades Foundation's trichotomy from quotation to theorem and closes the fourth road; failure documented is a finding.

C2 — the natural candidate: dissipative dynamics with zero-divisor jump operators. The framework supplies exactly one non-invertibility structure, and V measured it precisely: the anchored tunnel. C2 builds the Lindblad-type semigroup with jump operator L_A (anchor A on the locus), on the finite-mode system first, and asks the pre-registered questions (see P7 for the corrected, spectrum-valued form of the first): **(i)** does χ_{kin} deepen (depth-motion — the defining settling property, tested as a partial-order condition on the full mode-spectrum, not a scalar; the dynamics is non-unitary, so the scoped theorem permits it; whether it *does* is the test); **(ii)** what does it do to the wall sector's vacua manifold (the carrier verdict at dynamical level: pre-registered — the anchored dynamics transports the anchor's fiber-sign into the state's orientation datum, the dial turning *because history set it*, never spontaneously); **(iii)** the scalar-avoidance corollary at dynamical level (the kernel never touches the real units — does the dissipation therefore conserve the lone-scalar channel? A sharp, checkable question with measurement-stratum resonance). Instruments: the χ_{kin} -meter (symplectic spectrum; exists from Stage C of Phase 5), the vacua-manifold reader (V §3), both re-registered under P6 with their extensions to non-unitary dynamics explicitly checked — the Stage-C addendum of V applies with full force here.

C3 — the family question. If C2's candidate settles: is it distinguished within its class (anchor-averaged version, per V, is orientation-blind — is it also χ -static? Pre-registered by the $T_{\text{avg}} = 2 \cdot \text{Id}$ result: the democratic semigroup's motion, if any, is pure type-dynamics), and what selects the anchor remains exactly the open question — C3 documents what the candidate *cannot* decide, in V's carrier vocabulary.

4. Stage D — The anchor-supplier question in its own territory (gated on A-C; exploratory, strictest honesty clauses)

With the dossier, the wall verdict, and a working candidate: search the non-symplectic stratum — the only unaudited territory — for anything that supplies the anchor. Pre-priced outcomes: *a supplier with derived provenance* (the corpus's largest possible finding; extraordinary claims protocol applies — everything two-implemented, the Phase 4 provenance audit re-run on the new object before any announcement); *no supplier, with the search-space characterized* (the boundary theorem in strengthened form: the initial-condition verdict extended into the state stratum — pre-registered as the *expected* outcome, so that finding it is confirmation, not disappointment); *ill-posedness of the search itself* (documented, returned to the registry). Stage D cannot run before C's instruments exist and its honesty framing is proven in use.

5. Outcomes, pre-priced at program level

O1 — dossier + wall verdict + a settling candidate that moves χ with the carrier property confirmed dynamically: the state stratum acquires its first concrete dynamics, and the corpus's story becomes end-to-end executable. **O2** — the candidate fails to settle: the mechanism class narrows again; the failure characterized is the finding (the pattern of V honored). **O3** — B1 excludes the second wall: a named no-go; the e_R boundary becomes permanent structure. **O4** — D stays gated or lands on the boundary theorem: the program closes honestly on A-C, with the initial-condition verdict deepened rather than escaped. Every outcome produces a document.

6. Post-review amendments (round 1 — all five points accepted)

P1 (C2's branch choice, justified rather than silent). Foundation's trichotomy collapses, on inspection, into **two constructive branches**: the dissipative and open-system branches are one class (a Lindblad semigroup is an open system's reduced dynamics, by dilation), so the real fork is **Lindblad versus restriction**. The reviewer's point stands: a restriction map narrowing support over the scale line is arguably the more *literal* reading of "settling" — and the corpus even supplies its natural form (conditional expectation onto scale-truncated subalgebras, riding the half-sided structure). The order is therefore justified by **readiness, not naturalness**: the dissipative candidate's ingredients are already built, two-implemented, and orientation-characterized (V's L_A , T_A , P_kernel), and its instruments extend checkably (P3); the restriction candidate requires new objects. C2 runs dissipative-first; **C2' names the restriction candidate as the deferred second, not a dropped alternative**, activated if C2's candidate fails to settle or as the comparison once it succeeds. Depth-motion (χ) remains the *defined* settling signature per Foundation regardless of branch.

P2 (B1 given decidable test shapes — the parity criterion). "Forced" and "excluded" now have exact forms. B1's actual task: **determine, from the closed record alone, the sign of the mass structure at each of the scale line's two asymptotics**. Then parity decides: *same sign at both ends* + the one established crossing $\implies$ an even crossing count $\implies$ a second wall **forced**; *opposite signs* $\implies$ odd count, minimal one, with any structural monotonicity argument then **excluding** a second; *sign fixed at only one end or neither* $\implies$ **undetermined**, honestly, with the specific missing datum named. The stage is now a two-sign determination with a parity theorem behind it — either the record contains the two signs or it doesn't, and B1's report states which.

P3 (the Gaussianity gate — Stage C's precondition, per the reviewer's hold). The χ_{kin} -meter is well-defined along the candidate's flow only if the dynamics is Gaussian-preserving, which requires **jump operators linear in the mode operators**. The implementation is pinned accordingly: the anchor's

dissipation enters as the set of linear jump operators built from L_A 's matrix elements ($c_j = \sum_k (L_A)_{jk} a_k$), not as any quadratic one-body form. Under this choice the covariance matrix obeys the standard closed-form ODE and $\chi_{\text{kin}}(t)$ is well-defined (though no longer conserved — which is the point). **C2's first computation, before any settling question is asked:** verify Gaussianity-preservation and the covariance ODE on the finite-mode system, and re-register the χ_{kin} -meter and vacua reader under P6 with this extension explicit. No χ_{kin} -claim is admissible before this gate passes.

P4 (Stage D's effort clause, the reviewer's wording adopted). "Expected" governs how the null finding is *reported*, never how hard it is *looked for*: the search-space characterization accompanying a null result must meet the same completeness bar it would need if a positive result were expected.

P5 (the extraordinary-claims protocol, defined here since it gates the program's largest possible finding). Any candidate anchor-supplier triggers, in order: (i) full two-implementation of the object and its entire provenance chain; (ii) the Phase 4 sign-provenance audit re-run on the new object (derived versus convention versus gauge, by the standing instruments); (iii) pre-registration of the claim's falsifiers before any confirming computation is accepted as such; (iv) a dedicated hostile-review round on the claim alone, separate from any stage review; (v) entry into the corpus at [CANDIDATE] weight for one full review cycle before any upgrade. Precedents: the Phase 4 audit vocabulary and V's carrier-verdict process, both of which this protocol generalizes.

Freeze protocol: §§0–6 to the reviewer; the reviewer's hold was specifically on P2's criteria and P3's gate — both now concrete; confirmation on sight freezes, with Stage A (the dossier compilations) as the frozen program's first logged work.

7. Post-review amendments (round 2 — the χ -meter's target and form; numbered P7 to avoid collision with the standing-discipline P6 cited in §0 and §3)

P7 (the χ -meter's target quantity, and C2's test given its correct spectrum-valued, partial-order form). Two gaps, closed together since they're one fix. First: "the χ -meter" as written targets an unspecified χ — but C2's dynamics is dissipative, so its states can drift off the equilibrium spine entirely, where χ_{sector} and $\nu(\beta)$ (both folium/equilibrium-indexed) are not even defined. Becoming's own χ_{kin} — the symplectic spectrum evaluated directly on whatever covariance a state actually has, needing no folium, no equilibrium — is the only one of Becoming's four variants defined at every point C2's flow can reach. The χ -meter is pinned to χ_{kin} , by name, throughout.

Second: for a multi-mode system, χ_{kin} is not a scalar — it is the *full symplectic spectrum*, one eigenvalue per mode. Question (i) ("does it move χ ") and ques-

tion (ii) (“is its motion directed”) were two separate checks on an implicitly scalar quantity; on the correct, spectrum-valued χ_{kin} they merge into one well-posed test. Define $\chi_{\text{kin}}(t) \succeq \chi_{\text{kin}}(t')$ (component-wise partial order, in a fixed mode-basis) for $t > t'$ iff every mode’s own eigenvalue is weakly deeper at t than at t' . **Deepens:** $\chi_{\text{kin}}(t)$ is monotonically non-decreasing in this order for all t — every mode moves the same way, none reverses. **Wanders:** the order fails to hold at some pair $t > t'$ — some mode deepens while another shallows, or a later time regresses a mode past an earlier one. This is what “settling should settle, not wander” now means operationally, not metaphorically: a single spectrum-valued check, not two separately-motivated ones. C2’s four questions become three: (i) does χ_{kin} deepen in this sense (subsuming old (i)–(ii)); (ii) the vacua-manifold question (old (iii)); (iii) the scalar-avoidance corollary (old (iv)). P3’s Gaussianity gate is unchanged in substance — $\chi_{\text{kin}}(t)$ is exactly what the covariance ODE makes well-defined — but its own text now reads $\chi_{\text{kin}}(t)$ throughout, not bare $\chi(t)$.